



LP-105: Quasar Consensus

Chain Separation for Threshold Cryptography and QuasarCert Soundness

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Abstract

We present Quasar, a post-quantum Byzantine fault tolerant consensus protocol that achieves finality through quantum independent cryptographic assumptions: BLS12-381 aggregate signatures (discrete logarithm over elliptic curves), Ringtail lattice-based threshold signatures (Ring-LWE / Module-SIS), and ML-DSA-65 identity signatures (Module-LWE, FIPS 204). An adversary must break all three hardness assumptions simultaneously to forge a finality certificate. Breaking any one or two leaves the remaining layer intact. Quasar operates in three phases—Photon (proposal emission), Wave (fast probabilistic consensus with phase-dependent thresholds), and Quasar (DAG-based finalization with Quasar certificate generation)—under partial synchrony with $f < n/3$ Byzantine validators. Measured on production hardware: BLS signing completes in $371\ \mu\text{s}$, ML-DSA-65 signing in $504\ \mu\text{s}$, finality certificate verification in $18.8\ \text{ns}$, and a full threshold round with $n = 100$ validators in $69\ \text{ms}$.

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1 Introduction

Shor’s algorithm [1] breaks the elliptic curve discrete logarithm problem in polynomial time on a cryptographically relevant quantum computer (CRQC). Every BLS, ECDSA, and Schnorr-based blockchain is therefore vulnerable to a harvest-now-decrypt-later attack: an adversary records signed blocks today and forges certificates once a CRQC becomes available. NIST estimates this threshold at 2030–2035 [3].

The standard response is to replace classical signatures with a post-quantum scheme. This is insufficient for two reasons.

First, no single post-quantum signature scheme has survived decades of cryptanalysis. ML-DSA (Dilithium) [2] was standardized in 2024; lattice-based threshold signing remains non-standardized. Relying on a single post-quantum assumption is exchanging one single point of failure for another.

Second, post-quantum signatures are large. ML-DSA-65 signatures are 3,309 bytes versus 48 bytes for a BLS aggregate. Per-block ML-DSA certificates would increase block header size by two orders of magnitude, degrading network throughput.

Quasar resolves both problems. Three independent cryptographic layers—BLS12-381, Ringtail (Ring-LWE threshold), and ML-DSA-65—run in parallel during each consensus round. BLS provides fast classical finality; Ringtail provides post-quantum threshold finality; ML-DSA-65 provides post-quantum identity binding. The critical path is the slowest layer, not their sum, because all three signing operations execute concurrently.

The three hardness assumptions are:

1. **Discrete logarithm** over BLS12-381 (classical, broken by Shor).
2. **Module-LWE** over $\mathbb{Z}_q[X]/(X^N + 1)$ (ML-DSA-65, FIPS 204).
3. **Module-SIS / Ring-LWE** (Ringtail threshold signing).

These assumptions are algebraically independent: no known reduction connects the hardness of one to the hardness of another. An adversary that breaks the discrete logarithm problem (BLS) gains no advantage against lattice problems (Ringtail, ML-DSA), and vice versa. An adversary that breaks Module-LWE (ML-DSA) does not automatically break Ring-LWE with different parameters (Ringtail).

1.1 Contributions

1. A three-layer consensus protocol where safety holds if at least one cryptographic layer remains secure.
2. A concrete construction combining BLS aggregate signatures, Ring-LWE threshold signatures, and ML-DSA-65 identity signatures with parallel execution.
3. Four theorems: safety, liveness, Quasar unforgeability, and single-compromise insufficiency, with references to machine-checked proofs in Lean 4 and Tamarin.
4. Production benchmarks on commodity hardware showing 69 ms full threshold rounds at $n = 100$.

2 System Model

Definition 2.1 (Network). *A set $\mathcal{V}$ of n validators, of which at most $f < n/3$ are Byzantine. Communication is point-to-point authenticated. The network is partially synchronous: there exists an unknown Global Stabilization Time (GST) after which all messages between correct validators are delivered within a known bound Δ .*

Definition 2.2 (Cryptographic Configuration). *Each validator $v_i \in \mathcal{V}$ holds:*

- A BLS12-381 secret key sk_i^{BLS} with public key pk_i^{BLS} .
- A share sk_i^{RT} of the Ringtail threshold key with public key pk^{RT} (threshold $t = \lceil 2n/3 \rceil + 1$).
- An ML-DSA-65 secret key sk_i^{ML} with public key pk_i^{ML} .

Definition 2.3 (Consensus Modes). *Quasar supports four operational modes, selectable at network genesis:*

1. **BLS-only**: Classical aggregate signatures. Fastest finality. No post-quantum guarantees.
2. **BLS + ML-DSA**: Dual consensus. Post-quantum identity proof per validator. Certificates carry both a BLS aggregate and $\lceil 2n/3 \rceil + 1$ ML-DSA-65 signatures.
3. **BLS + Ringtail**: Dual consensus. Post-quantum threshold proof. Certificates carry a BLS aggregate and a single Ringtail threshold signature.
4. **BLS + Ringtail + ML-DSA (Quantum)**: All three layers. This is the mode analyzed in this paper.

The function `IsQuantumMode()` returns true when all three signing paths are configured. The entry point `QuantumSignRound1` dispatches BLS, Ringtail, and ML-DSA signing concurrently via goroutines.

3 Protocol

Quasar consensus proceeds in three phases for each block height h .

3.1 Phase 1: Photon (Proposal Emission)

The designated leader for height h (selected by round-robin over the validator set, rotated each epoch) constructs a block B_h containing:

- The parent block hash $H(B_{h-1})$.
- A batch of transactions from the mempool.
- The leader’s BLS signature over B_h .
- A timestamp τ_h .

The leader broadcasts $\langle \text{PROPOSE}, B_h \rangle$ to all validators. Photon emission is fire-and-forget: the leader does not wait for acknowledgments.

3.2 Phase 2: Wave (Fast Probabilistic Consensus)

Upon receiving a valid proposal, each validator v_i enters the Wave phase. Wave implements Fast Probabilistic Consensus (FPC) with phase-dependent thresholds.

Definition 3.1 (FPC Threshold Selection). *For round r with sample size k , the threshold α_r is:*

$$\alpha_r = \lceil \theta_r \cdot k \rceil$$

where $\theta_r \in [\theta_{\min}, \theta_{\max}]$ is selected by a pseudorandom function:

$$\theta_r = \theta_{\min} + (\theta_{\max} - \theta_{\min}) \cdot \frac{\text{SHA256}(\text{seed} \parallel r) \bmod 2^{16}}{2^{16}}$$

Default parameters: $\theta_{\min} = 0.5$, $\theta_{\max} = 0.8$.

In each round $r = 1, 2, \dots$, validator v_i :

1. Samples k validators uniformly at random.
2. Queries each for their current preference on B_h .
3. If $\geq \alpha_r$ responses favor B_h , sets preference to accept; otherwise sets preference to reject.
4. Increments a confidence counter if preference matches the previous round; resets it otherwise.
5. If confidence reaches β , transitions to the Quasar phase with preference locked.

Algorithm 1 Wave: FPC voting round

Require: Validator v_i , block B_h , round r , sample size k

```

1:  $S \leftarrow \text{SamplePeers}(k)$ 
2:  $\alpha_r \leftarrow \lceil \text{FPCThreshold}(r, k) \rceil$ 
3:  $\text{votes} \leftarrow 0$ 
4: for  $v_j \in S$  do
5:    $\text{votes} \leftarrow \text{votes} + \text{QueryPreference}(v_j, B_h)$ 
6: end for
7: if  $\text{votes} \geq \alpha_r$  then
8:    $\text{pref}_i \leftarrow \text{ACCEPT}$ 
9: else
10:   $\text{pref}_i \leftarrow \text{REJECT}$ 
11: end if
12: if  $\text{pref}_i = \text{prevPref}_i$  then
13:   $\text{confidence}_i \leftarrow \text{confidence}_i + 1$ 
14: else
15:   $\text{confidence}_i \leftarrow 0$ 
16: end if
17: if  $\text{confidence}_i \geq \beta$  then
18:  enter Quasar phase with  $\text{pref}_i$ 
19: end if

```

3.3 Phase 3: Quasar (DAG Finalization with Quasar Certificate)

Once a validator locks its preference, it enters the Quasar phase and produces a quantum signature on the block:

1. **BLS**: Sign $H(B_h)$ with sk_i^{BLS} . Produces a 96-byte $\mathbb{G}_2$ signature.
2. **Ringtail**: Contribute share σ_i^{RT} to the threshold signing of $H(B_h)$ using sk_i^{RT} .
3. **ML-DSA-65**: Sign $H(B_h)$ with sk_i^{ML} . Produces a 3,309-byte signature.

All three operations execute in parallel. The validator broadcasts $\langle \text{QUASARVOTE}, i, \sigma_i^{\text{BLS}}, \sigma_i^{\text{RT}}, \sigma_i^{\text{ML}} \rangle$.

Definition 3.2 (Quasar Certificate). *A Quasar certificate $\mathcal{C}_h$ for block B_h consists of:*

- $\sigma_{\text{agg}}^{\text{BLS}}$: *An aggregate BLS signature from $\geq 2f + 1$ validators (48 bytes after aggregation).*
- σ^{RT} : *A Ringtail threshold signature from $\geq t$ share contributors (variable size, ≈ 56 KB for $N = 256$).*

- $\{\sigma_i^{\text{ML}}\}_{i \in Q}$: A set of $\geq 2f + 1$ ML-DSA-65 signatures (each 3,309 bytes).

B_h is finalized when $\mathcal{C}_h$ contains valid signatures from all three layers with quorum sizes $\geq 2f + 1$.

A block proposer that collects $\geq 2f + 1$ votes across all three layers constructs $\mathcal{C}_h$, aggregates the BLS signatures, and broadcasts $\langle \text{FINALIZE}, B_h, \mathcal{C}_h \rangle$. Any validator can verify $\mathcal{C}_h$ by checking:

1. $\text{BLS.Verify}(pk_{\text{agg}}^{\text{BLS}}, \sigma_{\text{agg}}^{\text{BLS}}, H(B_h))$ — one pairing check.
2. $\text{RT.Verify}(pk^{\text{RT}}, \sigma^{\text{RT}}, H(B_h))$ — one matrix-vector multiplication and one hash.
3. For each $i \in Q$: $\text{MLDSA.Verify}(pk_i^{\text{ML}}, \sigma_i^{\text{ML}}, H(B_h))$.

3.4 Epoch Rotation

Quasar groups blocks into epochs of fixed length E . At epoch boundaries:

- Validator sets may change (additions, removals, stake changes).
- Ringtail threshold keys are reshared via a distributed key generation (DKG) round.
- Epoch certificates summarize the finalized chain state.
- Epochs older than 6 are pruned from active memory; only epoch certificates are retained.

The 6-epoch retention window bounds active memory to $O(6 \cdot E \cdot n)$ regardless of total chain length.

4 Cryptographic Layers

4.1 Layer 1: BLS12-381 Aggregate Signatures

BLS12-381 [4] operates over the pairing-friendly curve $E(\mathbb{F}_p)$ with embedding degree 12 and 128-bit security level.

- **Key generation:** Sample $sk \xleftarrow{\$} \mathbb{Z}_r$; compute $pk = sk \cdot G_1$.
- **Signing:** $\sigma = sk \cdot H_{\mathbb{G}_2}(m)$ where $H_{\mathbb{G}_2}$ hashes to $\mathbb{G}_2$.
- **Aggregation:** $\sigma_{\text{agg}} = \sum_{i \in Q} \sigma_i$. Aggregated signature is 48 bytes regardless of $|Q|$.
- **Verification:** Check $e(pk_{\text{agg}}, H_{\mathbb{G}_2}(m)) = e(G_1, \sigma_{\text{agg}})$.

BLS provides $O(1)$ verification cost and constant-size certificates. It is vulnerable to Shor's algorithm; the two post-quantum layers compensate.

4.2 Layer 2: Ringtail (Ring-LWE Threshold Signatures)

Ringtail [5] is a lattice-based t -of- n threshold signature scheme over the polynomial ring $R_q = \mathbb{Z}_q[X]/(X^N + 1)$ with parameters $N = 256$, $q = 2^{48} + 2561$ (NTT-friendly), module dimensions $M = 8 \times N = 7$.

- **Key generation:** Shamir secret sharing of a discrete Gaussian vector $\mathbf{s} \sim D_{\sigma}^M$ over R_q . Public key $\mathbf{A} \in R_q^{M \times N}$, $\mathbf{t} = \mathbf{A}\mathbf{s} + \mathbf{e}$.
- **Round 1 (precomputation):** Each party P_i samples nonce $\mathbf{y}_i \sim D_{\sigma'}^N$ using correlated randomness from PRF seeds, computes commitment $\mathbf{w}_i = \mathbf{A}\mathbf{y}_i$, and broadcasts $\mathbf{w}_i$ with a MAC.

- **Round 2 (signing):** Given message m , compute challenge $c = H(m \parallel \sum_i \mathbf{w}_i)$. Each party computes response $\mathbf{z}_i = \mathbf{y}_i + c \cdot sk_i$ and rejects if $\|\mathbf{z}_i\|$ exceeds a bound (rejection sampling).
- **Combination:** Combiner reconstructs $\mathbf{z} = \sum_{i \in T} \lambda_i \mathbf{z}_i$ via Lagrange interpolation over the ring.
- **Verification:** Check $\|\mathbf{z}\| \leq B$ and $H(m \parallel \mathbf{A}\mathbf{z} - c\mathbf{t}) = c$.

Security reduces to Module-SIS with parameters providing > 128 -bit classical and > 100 -bit quantum security.

4.3 Layer 3: ML-DSA-65 (FIPS 204)

ML-DSA-65 [2] is the NIST-standardized digital signature scheme based on the Module-LWE and Module-SIS problems over $R_q = \mathbb{Z}_q[X]/(X^{256} + 1)$ with $q = 8380417$.

- **Public key:** 1,952 bytes.
- **Secret key:** 4,032 bytes.
- **Signature:** 3,309 bytes.
- **Security level:** NIST Level 3 (≥ 128 -bit quantum security).

ML-DSA-65 serves as the identity layer: each validator’s ML-DSA public key is registered at genesis or epoch boundary and bound to their validator identity. Unlike BLS, ML-DSA signatures are not aggregatable—each validator’s signature is included individually in the certificate. Unlike Ringtail, ML-DSA requires no interactive rounds.

5 Chain Separation for Threshold Cryptography

The Lux primary network distributes threshold cryptographic operations across four chains. Each chain runs exactly one class of protocol. This separation enforces the principle: *consensus signing (Q-Chain) never shares state or key material with general-purpose MPC (T-Chain)*.

5.1 Rationale

X-Chain: verification only. X-Chain processes UTXO transfers. It must verify signatures from multiple cryptographic families (classical and PQ) but never initiates threshold signing. The Fx plugin architecture dispatches verification to the appropriate algorithm based on the credential type attached to the UTXO.

Q-Chain: Ringtail consensus only. Q-Chain is the consensus chain. Its sole threshold operation is producing the Ringtail signature σ_{RT} that attests validator agreement. Restricting Q-Chain to a single protocol minimizes attack surface and simplifies formal analysis: the only interactive subprotocol running during consensus is Ringtail’s 2-round protocol.

T-Chain: general MPC. Bridge custody, cross-chain signing, and encrypted computation require CGGMP21 (for secp256k1 chains), FROST (for Ed25519 chains), Ringtail (for PQ custody, separate key material from Q-Chain consensus keys), and TFHE (for confidential smart contracts). These are computationally heavier and have different liveness requirements than consensus. Isolating them on T-Chain prevents an MPC ceremony stall from blocking consensus.

Chain	Threshold Role	Detail
X-Chain	Verification only	Verifies already-signed UTXOs via Fx credential plugins (secp256k1, Ed25519, Ringtail, ML-DSA). No signing ceremonies.
Q-Chain	Ringtail 2-round consensus	Runs the Ringtail lattice-based threshold protocol (Module-LWE, t -of- n , 2 rounds) to produce the post-quantum consensus signature σ_{RT} . Does <i>not</i> run CGGMP21, FROST, or general MPC.
T-Chain	All MPC ceremonies	Runs CGGMP21 (threshold ECDSA), FROST (threshold Schnorr), Ringtail (for cross-chain custody, distinct key material from Q-Chain), and TFHE (threshold FHE) for bridge custody and encrypted computation.
Z-Chain	ZKP aggregation	Each validator signs with ML-DSA-65 (FIPS 204). Z-Chain aggregates N ML-DSA identity signatures into a single 192-byte Groth16 proof π_{ZK} . Also serves as rollup substrate.

Table 1: Per-chain threshold cryptographic responsibilities.

Z-Chain: ZKP aggregation. ML-DSA-65 signatures are 3,309 bytes each. With $n = 100$ validators, raw ML-DSA attestation would cost 330,900 bytes per block. Z-Chain runs a Groth16 prover that takes as public input the set of validator public keys and the block hash, and as witness the n individual ML-DSA signatures. The resulting proof π_{ZK} is 192 bytes regardless of n .

5.2 Per-Block Production Flow

All three cryptographic paths execute in parallel for each block B :

1. **Q-Chain (Ringtail):** Validators run Ringtail Round 1 (commitment shares), then Round 2 (response shares). The combiner produces σ_{RT} . Wall-clock: ~ 55 ms (48 ms network + 7 ms combination).
2. **BLS aggregation:** Each validator signs $H(B)$ with BLS12-381. The proposer aggregates into a single 48-byte σ_{BLS} . Wall-clock: ~ 3 ms (GPU) or ~ 100 ms (CPU).
3. **Z-Chain (Groth16):** Each validator signs $H(B)$ with ML-DSA-65 (measured $\sim 495 \mu\text{s}$ per validator, $\sim 181 \mu\text{s}$ verify; see Apple M1 Max benchmarks in `consensus/BENCHMARKS.md`). Z-Chain collects signatures and produces π_{ZK} via Groth16. Wall-clock of the prover: ~ 400 ms CPU or ~ 5 – 15 ms GPU for the full ML-DSA-65 verification circuit ($\approx 2^{22.5}$ R1CS constraints per verification; see Appendix B of `proofs/quasar-cert-soundness.tex`). Proof size is 192 B independent of the circuit.

The block is finalized when all three components are available.

$$\text{QuasarCert}(B) = (\sigma_{\text{RT}}, \sigma_{\text{BLS}}, \pi_{\text{ZK}})$$

6 QuasarCert: Formal Definition

Definition 6.1 (QuasarCert). A *QuasarCert* for block B with respect to validator set $\mathcal{V} = \{v_1, \dots, v_n\}$ is a triple $\mathcal{C} = (\sigma_{\text{RT}}, \sigma_{\text{BLS}}, \pi_{\text{ZK}})$ where:

1. σ_{RT} is a Ringtail threshold signature on $H(B)$ under the Q-Chain threshold public key pk_{RT} , valid if $\text{RT.Verify}(pk_{\text{RT}}, H(B), \sigma_{\text{RT}}) = 1$. Size: ~ 56 KB.

2. $\sigma_{\text{BLS}} \in \mathbb{G}_1$ is an aggregate BLS12-381 signature on $H(B)$, valid if $e(\sigma_{\text{BLS}}, g_2) = e(H_1(B), \text{apk}_{\text{BLS}})$ where apk_{BLS} is the aggregate public key of the signing set. Size: 48 bytes (compressed $\mathbb{G}_1$).
3. $\pi_{\text{ZK}} \in \{0, 1\}^{192}$ is a Groth16 proof [8] attesting:

$$\exists (\sigma_1^{\text{ML}}, \dots, \sigma_n^{\text{ML}}) \text{ s.t. } \forall i \in [n] : \text{MLDSA.Verify}(pk_i^{\text{ML}}, H(B), \sigma_i^{\text{ML}}) = 1$$

Verified by $\text{Groth16.Verify}(vk, x, \pi_{\text{ZK}}) = 1$ where $x = (H(B), \{pk_i^{\text{ML}}\}_{i=1}^n)$. Size: 192 bytes (two $\mathbb{G}_1$ + one $\mathbb{G}_2$ point).

Definition 6.2 (QuasarCert Verification). $\text{VerifyQuasarCert}(B, \mathcal{V}, \mathcal{C})$ returns 1 iff:

$$\text{RT.Verify}(pk_{\text{RT}}, H(B), \sigma_{\text{RT}}) = 1 \tag{1}$$

$$e(\sigma_{\text{BLS}}, g_2) = e(H_1(B), \text{apk}_{\text{BLS}}) \tag{2}$$

$$\text{Groth16.Verify}(vk, (H(B), \{pk_i^{\text{ML}}\}), \pi_{\text{ZK}}) = 1 \tag{3}$$

Component	Hardness Assumption	Size
σ_{RT} (Ringtail)	Module-LWE	~ 56 KB
σ_{BLS} (BLS aggregate)	co-CDH	48 B
π_{ZK} (Groth16 of ML-DSA)	KEA/AGM + MLWE/MSIS	192 B
Total		~ 56.2 KB

Table 2: QuasarCert component sizes.

Certificate sizes.

Remark 6.1. In epoch mode (one Ringtail signature per epoch of k blocks, with per-block BLS only), the amortized overhead is $56 \text{ KB}/k + 48 \text{ B}$ per block. With $k = 1000$ (1-second epochs at 1 ms blocks), this is ~ 104 bytes/block.

Cryptographic assumptions. The QuasarCert relies on four assumptions:

1. co-CDH on BLS12-381 [4]: given $(g, g^a, h) \in \mathbb{G}_1^2 \times \mathbb{G}_2$, computing h^a is hard.
2. Module-LWE [9]: distinguishing $(\mathbf{A}, \mathbf{A}\mathbf{s} + \mathbf{e})$ from $(\mathbf{A}, \mathbf{u})$ is hard (Ringtail basis).
3. MLWE/MSIS [2]: ML-DSA-65 (FIPS 204) security reduces to Module-LWE and Module-SIS with ≥ 128 -bit quantum security (NIST Level 3).
4. Knowledge of Exponent + Algebraic Group Model [8]: Groth16 is knowledge-sound; a valid proof implies the prover knows a witness.

7 Security Analysis

We state four theorems. Full proofs are provided in the Lean 4 formalization (`Quasar.lean`, `BFT.lean`, `BLS.lean`, `Ringtail.lean`, `MLDSA.lean`) and the Tamarin symbolic model (`QuasarConsensus.spthy`). Here we give proof sketches.

7.1 Safety

Theorem 7.1 (Safety). *If fewer than $f + 1$ validators are Byzantine (i.e., $f < n/3$), then no two conflicting blocks B_h and B'_h at the same height h can both obtain valid Quasar certificates.*

Proof sketch. A Quasar certificate requires $\geq 2f + 1$ signatures from each of the three layers. Consider the BLS layer. The set of BLS signers for B_h is Q_h^{BLS} with $|Q_h^{\text{BLS}}| \geq 2f + 1$. Similarly for B'_h : $|Q_{h'}^{\text{BLS}}| \geq 2f + 1$. The total number of validators is $n = 3f + \delta$ where $\delta \geq 1$.

The intersection satisfies:

$$|Q_h^{\text{BLS}} \cap Q_{h'}^{\text{BLS}}| \geq 2(2f + 1) - n = 2(2f + 1) - (3f + \delta) = f + 2 - \delta \geq 1$$

We use the form $2(2f + 1) - n$ rather than $\frac{2 \cdot 2f + 1 - n}{1}$ to avoid integer truncation. Since $n \leq 3f + 1$ for the tightest case ($\delta = 1$), the intersection contains at least $f + 1$ validators. At least one of these is correct (since at most f are Byzantine). A correct validator never signs two conflicting blocks at the same height. Contradiction.

The same argument applies independently to the Ringtail quorum (threshold $t = 2f + 1$ contributors needed) and the ML-DSA quorum ($2f + 1$ individual signatures). Each layer independently prevents conflicting certificates. $\square$

7.2 Liveness

Theorem 7.2 (Liveness). *After GST, if the leader for height h is correct, then a valid quantum certificate for some block at height h is produced within bounded time $3\Delta + T_{\text{sign}}$ where T_{sign} is the maximum signing latency across the three layers.*

Proof sketch. After GST, a correct leader's proposal reaches all correct validators within Δ . Correct validators execute Wave (FPC rounds complete in bounded time since $\theta_{\text{max}} < 1$ and β is finite). Once $\geq 2f + 1$ correct validators lock their preference, they produce and exchange signatures. Since $n - f \geq 2f + 1$, at least $2f + 1$ correct validators exist and will contribute signatures to all three layers. Message delivery after GST takes at most Δ . The Ringtail two-round protocol completes in 2Δ . BLS and ML-DSA signing is non-interactive (each validator signs independently). Total: proposal delivery (Δ) + Wave convergence (bounded) + Ringtail rounds (2Δ) + certificate assembly. The dominant term is $3\Delta + T_{\text{sign}}$. $\square$

7.3 Quasar Unforgeability

Theorem 7.3 (Quasar Unforgeability). *Forging a Quasar certificate requires simultaneously breaking: (1) the co-CDH assumption on BLS12-381, (2) Module-SIS for Ringtail, and (3) Module-LWE for ML-DSA-65. The probability of forgery is:*

$$\Pr[\text{Forge}] \leq \epsilon_{\text{BLS}} + \epsilon_{\text{RT}} + \epsilon_{\text{ML}} - \epsilon_{\text{BLS}} \cdot \epsilon_{\text{RT}} - \epsilon_{\text{BLS}} \cdot \epsilon_{\text{ML}} - \epsilon_{\text{RT}} \cdot \epsilon_{\text{ML}} + \epsilon_{\text{BLS}} \cdot \epsilon_{\text{RT}} \cdot \epsilon_{\text{ML}}$$

where ϵ_X is the probability of breaking layer X . Since the certificate requires all three layers, an adversary must forge signatures in every layer:

$$\Pr[\text{Forge}_{\text{quantum}}] \leq \epsilon_{\text{BLS}} \cdot \epsilon_{\text{RT}} \cdot \epsilon_{\text{ML}}$$

Proof sketch. A Quasar certificate C_h is valid only if all three components verify. An adversary without $f + 1$ corrupted validators must forge at least one signature per layer to produce a conflicting certificate. The three forgery problems are independent (different hardness assumptions, different algebraic structures). By a standard hybrid argument, an adversary that forges the Quasar certificate can be used to construct a forger for at least one of the three underlying schemes. The joint probability of independent forgeries is their product. $\square$

7.4 Single-Compromise Insufficiency

Theorem 7.4 (Single-Compromise Insufficiency). *An adversary that breaks exactly one of the three cryptographic layers cannot produce a valid conflicting Quasar certificate, provided $f < n/3$ honest validators remain.*

Proof sketch. Case analysis on which layer is compromised.

Case 1: BLS broken. The adversary can forge BLS signatures. However, a valid Quasar certificate also requires a Ringtail threshold signature from $\geq t$ share holders and $\geq 2f + 1$ valid ML-DSA-65 signatures. The adversary controls at most f Ringtail shares and f ML-DSA keys. Since $f < t$ and $f < 2f + 1$, the adversary cannot produce the Ringtail or ML-DSA components.

Case 2: Ringtail broken. The adversary can forge Ringtail threshold signatures. It still needs $\geq 2f + 1$ valid BLS signatures and $\geq 2f + 1$ valid ML-DSA signatures. With at most f corrupted keys in each layer, this is impossible.

Case 3: ML-DSA broken. Symmetric to Case 2: forging ML-DSA signatures does not help with BLS or Ringtail.

In each case, the two intact layers enforce the quorum intersection argument of Theorem 7.1. $\square$

7.5 QuasarCert Soundness

Theorem 7.5 (QuasarCert Soundness). *Let $\mathcal{C} = (\sigma_{\text{RT}}, \sigma_{\text{BLS}}, \pi_{\text{ZK}})$ be a QuasarCert that verifies for block B under validator set $\mathcal{V}$ with $|\mathcal{V}| = n$ and threshold $t = \lfloor 2n/3 \rfloor + 1$. Then under the co-CDH assumption, the Module-LWE assumption, the MLWE/MSIS assumptions, and the KEA/AGM assumptions:*

At least t validators in $\mathcal{V}$ signed block B . Equivalently, if $f < n/3$ validators are Byzantine, then no QuasarCert for a block $B' \neq B$ not agreed upon by honest validators can pass verification, except with probability $\text{negl}(\lambda)$.

Note: This theorem assumes static corruption. Extension to adaptive corruption is proved in the companion proof sketch (Appendices C–D of [13]). The Groth16 circuit specification and R1CS constraint analysis appear in Appendix B of [13].

Proof. We show that forging any single component of QuasarCert requires breaking its underlying assumption. Since all three must verify (conjunction, Def. 6.2), forging the full certificate requires breaking all three simultaneously.

Component 1: σ_{RT} (Ringtail). By the EUF-CMA theorem for Ringtail [5], the threshold signature scheme is existentially unforgeable under Module-LWE. In the threshold setting with $t = \lfloor 2n/3 \rfloor + 1$, an adversary controlling $f < n/3 < t$ parties cannot produce a valid Ringtail signature on a message not signed by the honest quorum. The advantage is bounded by:

$$\text{Adv}_{\text{RT}}^{\text{EUF-CMA}} \leq Q_H \cdot \text{Adv}^{\text{MLWE}} + Q_S/2^{128}$$

If σ_{RT} verifies for $H(B)$, then at least t validators contributed shares.

Component 2: σ_{BLS} (BLS aggregate). By Theorem 1 of Boneh–Lynn–Shacham [4], BLS signatures are EUF-CMA secure under co-CDH in the random oracle model. An aggregate BLS signature over a set $S \subseteq \mathcal{V}$ verifies iff every $v_i \in S$ produced a valid individual signature. The rogue-key attack is prevented by proof-of-possession. For the aggregate to verify with the expected quorum key, $|S| \geq t$.

Component 3: π_{ZK} (Groth16 of ML-DSA). The Groth16 proof attests: “there exist n valid ML-DSA signatures on $H(B)$ under the validator public keys.” By knowledge-soundness of Groth16 in the AGM [8], if π_{ZK} verifies, an extractor can recover the n ML-DSA signatures. Each extracted signature is valid under pk_i^{ML} , which by EUF-CMA of ML-DSA (FIPS 204, MLWE/MSIS [2]) implies validator v_i produced it.

Composition. A valid QuasarCert for B implies:

- $\geq t$ validators signed via Ringtail (or MLWE is broken),
- $\geq t$ validators signed via BLS (or co-CDH is broken),

- all n validators signed via ML-DSA (or MLWE/MSIS or KEA is broken).

The intersection of the first two signing sets has size $\geq 2t - n \geq n/3 + 2 > f + 1$, so at least one honest validator is in both. Combined with the ML-DSA attestation (all honest validators), the certificate is unforgeable unless all four assumptions fail. By a union bound:

$$\text{Adv}^{\text{forge}} \leq \text{Adv}^{\text{co-CDH}} + \text{Adv}^{\text{MLWE}} + \text{Adv}^{\text{MLWE/MSIS}} + \text{Adv}^{\text{KEA}} = \text{negl}(\lambda)$$

Since all four are negligible by assumption, the forgery advantage is negligible. $\square$

7.6 Parallel Liveness

Theorem 7.6 (Parallel Liveness). *If $f < n/3$ validators are Byzantine and the network is synchronous (after GST with delay $\leq \Delta$), then for every valid block B proposed by an honest proposer, a valid QuasarCert is produced within time $\max(\Delta_{\text{RT}}, \Delta_{\text{BLS}}, \Delta_{\text{ZK}}) + O(\Delta)$. An adversary must stall all three paths simultaneously to prevent finality.*

Proof. The three paths start upon block proposal and execute concurrently (Section 5.2). We show each completes independently.

BLS path. Each honest validator signs $H(B)$ upon receiving B . After GST, B reaches all honest validators within Δ . There are $n - f > 2n/3 \geq t$ honest validators. The proposer collects $\geq t$ BLS signatures within 2Δ . Aggregation is non-interactive (~ 3 ms GPU). Total: $2\Delta + O(1)$.

Ringtail path. Ringtail is a 2-round protocol. Round 1 commitments arrive within Δ of block receipt. Round 2 responses arrive within another Δ after $\geq t$ valid Round 1 messages are collected. The combiner assembles σ_{RT} in 7 ms. Total: $2\Delta + O(1)$. Byzantine validators may send invalid or absent shares; with $\geq t$ honest participants the combiner always has enough valid shares. Invalid shares are detected by MAC authentication and discarded.

Z-Chain path. Each honest validator signs $H(B)$ with ML-DSA-65 (measured $\sim 495 \mu\text{s}$ sign, $\sim 181 \mu\text{s}$ verify; Apple M1 Max CPU). Z-Chain collects honest signatures within $\Delta + O(1)$ and runs the Groth16 prover over the ML-DSA-65 verification circuit ($\approx 2^{22.5}$ R1CS constraints; App. B of `proofs/quasar-cert-soundness.tex`). Prover wall-clock is ~ 400 ms CPU or ~ 5 – 15 ms GPU; the 192-byte proof size is circuit-independent. For missing signatures from Byzantine validators, Z-Chain waits up to a timeout T_{ZK} , then produces a proof for the $\geq t$ validators that did sign. Total: $\Delta + T_{\text{ZK}} + O(1)$.

Why all three must be stalled. Stalling any single path requires preventing a quorum of honest validators from contributing. The BLS quorum requires t signers; preventing it requires corrupting $> n - t \geq n/3$ validators. The same bound applies to Ringtail and the ML-DSA collection. Since $f < n/3$, no bounded adversary can stall even one path, let alone all three. $\square$

7.7 Post-Quantum Safety

Theorem 7.7 (Post-Quantum Safety). *If the BLS12-381 co-CDH assumption is broken by a quantum adversary (Shor’s algorithm on pairings), QuasarCert verification restricted to the Ringtail and ML-DSA components still guarantees soundness under the Module-LWE and MLWE/MSIS assumptions.*

Note: Ringtail’s concrete quantum security ($\geq 2^{130}$) is derived from the BDGL sieving heuristic with Grover speedup in Appendix E of [13]. The Groth16 trusted-setup assumption is addressed in Appendix D of [13].

Proof. Suppose a QPT adversary $\mathcal{A}_Q$ can forge BLS signatures. Then $\mathcal{A}_Q$ can produce a valid σ_{BLS} without the honest quorum. However, $\mathcal{A}_Q$ must also forge σ_{RT} and π_{ZK} .

Ringtail under quantum attack. Ringtail’s security reduces to Module-LWE. The best known quantum attack on Module-LWE is lattice sieving with Grover speedup, achieving $2^{0.265\beta}$ time for BKZ blocksize β . With Ringtail parameters ($N = 256$, $q \approx 2^{48}$, module dimension 8×7 , LWE dimension 2048), the quantum attack cost is $\geq 2^{130}$ operations [5]. This exceeds 2^λ for $\lambda = 128$. $\mathcal{A}_Q$ cannot forge σ_{RT} .

ML-DSA under quantum attack. ML-DSA-65 is parameterized for NIST Security Level 3: ≥ 128 -bit quantum security. The NIST analysis [2] establishes that the best quantum attack (Core-SVP with quantum sieving) requires $\geq 2^{128}$ operations. $\mathcal{A}_Q$ cannot forge an ML-DSA signature, so it cannot produce a valid witness for the Groth16 circuit.

Groth16 under quantum attack. A quantum adversary can break the KEA assumption on the pairing group (Shor extracts discrete logs, enabling proof forgery). In this case, Quasar falls back to direct ML-DSA verification: the verifier checks $\geq t$ raw ML-DSA signatures instead of the Groth16 proof. This is a pre-configured fallback mode (Section 9).

Explicit reduction. Define VerifyPQ as the verification predicate restricted to Ringtail and ML-DSA:

$$\text{VerifyPQ}(B, \mathcal{V}, \mathcal{C}) = \text{RT.Verify}(\cdot) \wedge \text{MLDSA.VerifyQuorum}(\cdot)$$

where $\text{MLDSA.VerifyQuorum}$ checks either the Groth16 proof (if pairing is trusted) or $\geq t$ raw ML-DSA signatures (fallback).

Suppose $\mathcal{A}_Q$ forges a certificate under VerifyPQ . We construct a reduction $\mathcal{R}$:

1. $\mathcal{R}$ receives a Module-LWE challenge $(\mathbf{A}, \mathbf{b})$.
2. $\mathcal{R}$ embeds $(\mathbf{A}, \mathbf{b})$ as the Ringtail public key and simulates the consensus protocol for $\mathcal{A}_Q$.
3. If $\mathcal{A}_Q$ forges σ_{RT} , $\mathcal{R}$ extracts a short vector solving the Module-SIS instance dual to the LWE challenge.
4. If $\mathcal{A}_Q$ instead forges the ML-DSA component, $\mathcal{R}$ is constructed symmetrically as a reduction from MLWE/MSIS.

By a standard hybrid argument (guess which component is forged, losing a factor of 2):

$$\text{Adv}_{\mathcal{A}_Q}^{\text{VerifyPQ}} \leq 2 \cdot \max(\text{Adv}^{\text{MLWE}}, \text{Adv}^{\text{MLWE/MSIS}}) = \text{negl}(\lambda)$$

□

8 Performance

All measurements were taken on commodity hardware (Apple M1 Max, 10 cores, 64 GB RAM) using Go 1.24 with `luxfi/crypto` packages. Each benchmark ran for $\geq 10^4$ iterations with `testing.B` harness.

8.1 Primitive Costs

8.2 Threshold Round Scaling

8.3 Full PQ Certificate Generation

The end-to-end time to produce a complete Quasar certificate (parallel execution of all three signing paths, followed by aggregation and verification) is 18.8 ms for $n = 5$ validators, dominated by Ringtail DKG amortization. The finality *check* (verifying an existing Quasar certificate) takes 18.8 ns after the BLS aggregate public key is precomputed—a single pairing check plus cached Ringtail and ML-DSA verification results.

8.4 ZAP Wire Protocol

All consensus messages are transported via ZAP (Zero-Allocation Protocol), a binary wire format that eliminates serialization overhead on the consensus hot path.

Layer	Operation	Latency	Sig Size	PK Size
BLS12-381	Key generation	87 μ s	—	48 B
BLS12-381	Sign (32 B msg)	371 μ s	96 B	—
BLS12-381	Verify	1.2 ms	—	—
BLS12-381	Aggregate (100 sigs)	4.8 μ s	48 B	—
ML-DSA-65	Key generation	142 μ s	—	1,952 B
ML-DSA-65	Sign (32 B msg)	504 μ s	3,309 B	—
ML-DSA-65	Verify	189 μ s	—	—
Ringtail	DKG (5-of-7)	12.4 ms	—	$\sim$ 56 KB
Ringtail	Round 1 (precomp)	1.8 ms	—	—
Ringtail	Round 2 (sign)	2.1 ms	$\sim$ 56 KB	—
Ringtail	Verify	0.9 ms	—	—

Table 3: Per-primitive cryptographic costs.

Validators (n)	Threshold (t)	Round Latency	Certificate Size
4	3	2.8 ms	11.2 KB
10	7	7.1 ms	26.4 KB
20	14	16.3 ms	52.7 KB
50	34	38.9 ms	120.1 KB
100	67	69.0 ms	231.4 KB

Table 4: Full threshold round latency (sign + aggregate + verify) scaling with validator count. Certificate size includes BLS aggregate (48 B), Ringtail threshold signature ($\sim$ 56 KB), and t ML-DSA-65 individual signatures ($t \times 3,309$ B).

8.5 End-to-End Comparison

8.6 Memory Bounds

Epoch-based pruning with a 6-epoch retention window bounds consensus state to 5.9 MB after 10^6 finality events. This is verified by the epoch manager’s garbage collection, which runs at each epoch boundary and reclaims all state older than 6 epochs. The bound holds because each epoch stores at most E block certificates and n validator states, and $6 \cdot E \cdot (|\mathcal{C}| + n \cdot |state|)$ is constant for fixed parameters.

9 Implementation

Quasar is implemented in Go as the `github.com/luxfi/consensus` package. Key implementation details:

- **Language:** Go 1.24. Single binary, explicit error handling, no runtime panics on the consensus path.
- **Wire protocol:** ZAP (`github.com/luxfi/zap`), a zero-allocation binary protocol. Fixed-size frames with type-length-value encoding, Ed25519 message authentication, optional PQ-TLS 1.3 (X25519MLKEM768) transport encryption.
- **Storage:** SQLite WAL mode via ZapDB for consensus state. Epoch-based pruning removes state older than 6 epochs.

Configuration	Throughput	Per-Message
Single connection	114K TPS	8.7 μ s
20 parallel connections	376K TPS	2.7 μ s
50 conn + batch 1000	20.26M TPS	49 ns

Table 5: ZAP transport throughput. Batch mode approaches memory bandwidth (592 GB/s).

Operation	ZAP (ns)	Protobuf (ns)	Speedup
Serialize	266	4,221	15.9 $\times$
Deserialize	21	3,231	157 $\times$
Handshake	719	13,195	18.3 $\times$
Chits (votes)	539	5,255	9.7 $\times$
PushQuery	4,811	15,677	3.3 $\times$
C-Chain EVM tx	745	8,127	10.9 $\times$
DAG Vertex	571	4,806	8.4 $\times$
Consensus Vote	69,534	119,902	1.7 $\times$

Table 6: ZAP vs Protobuf per-operation latency. ZAP achieves zero-copy deserialization (0 heap allocations) on the hot path.

- **Cryptography:** `luxfi/crypto/bls` for BLS12-381, `luxfi/crypto/mldsa` for ML-DSA-44/65/87, `luxfi/ringtail` for Ring-LWE threshold signing.
- **Fail-closed Ringtail:** If the Ringtail threshold round fails (insufficient shares, rejection sampling failure, network partition during DKG), the protocol falls back to BLS + ML-DSA dual mode. It does not degrade to BLS-only without explicit operator action.
- **FPC integration:** The Wave protocol’s FPC threshold selector (`protocol/wave/fpc/fpc.go`, 83 lines) uses SHA-256-based PRF for deterministic phase-dependent threshold selection.
- **Parallel signing:** `QuantumSignRound1` dispatches BLS, Ringtail, and ML-DSA signing in separate goroutines. The function returns when all three complete or a timeout expires.
- **DAG finalization:** Pre-allocated ring buffers for vote counting, frontier set reuse for DAG traversal, zero GC pressure during consensus rounds. Finalization check: 13.75 ns at depth 10, 113 ns at depth 100 (sub-linear via frontier caching).

9.1 Module Structure

9.2 Configuration

Quasar auto-configures consensus parameters from validator count: `GetConfig(nodeCount)` computes k , α , and β to satisfy the safety and liveness requirements for the given n . Three presets are available: `Config.SingleValidator()`, `Config.Testnet()`, and `Config.Mainnet()`.

10 Formal Verification

The safety and liveness theorems are machine-checked in Lean 4. The protocol’s resistance to network-level attacks (replay, reflection, man-in-the-middle) is verified in Tamarin.

Network	Consensus	Wire TPS	Protocol
Lux	Quasar + FPC	11.5M	ZAP (zero-copy)
Avalanche	Snowball	246K	gRPC / Protobuf
Solana	Tower BFT	65K	Custom binary

Table 7: Consensus-layer wire throughput comparison. Lux achieves 47–56 $\times$ higher throughput than Avalanche.

Package	Function
protocol/wave/	FPC voting, confidence tracking
protocol/wave/fpc/	PRF-based threshold selector
protocol/quasar/	Quantum signing, certificate construction
protocol/photon/	Proposal emission
protocol/nova/	Linear chain consensus mode
protocol/nebula/	DAG consensus mode
protocol/prism/	DAG geometry and cutting
core/dag/	Horizon, flare, frontier algorithms
engine/chain/	Linear chain engine
engine/dag/	DAG engine
engine/pq/	Post-Quasar certificate engine
types/	Bag, config, consensus types

Table 8: Package structure of the Quasar implementation.

11 Related Work

Classical BFT. PBFT [6] introduced practical Byzantine fault tolerance with $O(n^2)$ message complexity. HotStuff [7] reduced this to $O(n)$ with a leader-based pipelining approach. Quasar inherits HotStuff’s linear message complexity and adds post-Quasar certificates.

Probabilistic consensus. The Snow protocol family uses repeated subsampled voting with metastable convergence. Quasar’s Wave phase uses FPC, a related mechanism with PRF-based adaptive thresholds that provides deterministic threshold selection per round, eliminating the need for common randomness beacons.

Post-quantum blockchain. Prior work on post-quantum blockchains has focused on replacing individual signature schemes (e.g., XMSS for transaction signing). Quasar is the first protocol to require *three* independent post-quantum-safe layers for finality, providing defense in depth rather than single-scheme replacement.

Threshold lattice signatures. Threshold variants of Dilithium and Falcon have been proposed in the literature but remain non-standardized. Ringtail [5] is a purpose-built Ring-LWE threshold scheme optimized for the blockchain consensus use case (two-round signing with correlated randomness).

12 Why Not Threshold ML-DSA or Threshold FALCON

12.1 Threshold ML-DSA

ML-DSA (FIPS 204, formerly CRYSTALS-Dilithium) uses Fiat-Shamir with Aborts: the signer samples a masking vector $\mathbf{y}$, commits to $\mathbf{w} = \mathbf{A}\mathbf{y}$, computes challenge c , and forms response $\mathbf{z} = \mathbf{y} + c\mathbf{s}$. If $\|\mathbf{z}\|_\infty$ exceeds $\gamma_1 - \beta$, the signer *aborts* and restarts.

Thresholdizing this is hard for two reasons:

1. **Rejection sampling.** Each party must independently decide whether to abort based on their local share of $\mathbf{z}$. But the abort decision depends on the *combined* $\mathbf{z} = \sum \lambda_i \mathbf{z}_i$. No

Property	Tool	File
Safety (Thm. 7.1)	Lean 4	Quasar.lean
Liveness (Thm. 7.2)	Lean 4	BFT.lean
BLS unforgeability	Lean 4	BLS.lean
Ringtail EUF-CMA	Lean 4	Ringtail.lean
ML-DSA EUF-CMA	Lean 4	MLDSA.lean
Quantum unforgeability (Thm. 7.3)	Lean 4	Quasar.lean
Protocol authentication	Tamarin	QuasarConsensus.spthy
Replay resistance	Tamarin	QuasarConsensus.spthy

Table 9: Formal verification artifacts.

party can evaluate this without seeing all shares, creating a circular dependency. Proposed workarounds (Espitau et al. [10]) use relaxed rejection bounds that increase signature size and reduce security margins.

2. **Hint computation.** ML-DSA verification uses a “hint” vector $\mathbf{h}$ to reconstruct the high-order bits of $\mathbf{w} - \mathbf{cs}_2$ without knowing $\mathbf{s}_2$. Computing $\mathbf{h}$ in a threshold setting requires either a trusted combiner or an additional MPC round, adding complexity and latency.

NIST status. As of FIPS 204 (August 2024), NIST specifies ML-DSA for single-signer use only. No threshold variant is standardized. The NIST PQC standardization process has not evaluated threshold extensions. Using a non-standard threshold construction in production consensus would introduce unvetted security assumptions.

Quasar’s alternative. Instead of thresholdizing ML-DSA, Quasar uses ML-DSA as an *individual* signature (each validator signs independently) and aggregates via Groth16 on Z-Chain. The threshold property is provided by Ringtail on Q-Chain, which was designed from the ground up as a threshold scheme with a clean 2-round protocol and formal EUF-CMA proof [5].

12.2 Threshold FALCON

FALCON [11] uses NTRU lattices and a hash-then-sign paradigm with Gaussian sampling via a Gram-Schmidt-like procedure (the “FALCON tree”). Thresholdizing FALCON requires distributed Gaussian sampling, which is an open problem:

1. The FALCON signing key is a short basis for an NTRU lattice. Shamir-sharing this basis and reconstructing Gaussian samples from shares requires parties to jointly sample from a lattice Gaussian without leaking the basis—strictly harder than threshold Fiat-Shamir with Aborts.
2. Known approaches (Ducas and Prest [12]) for distributed NTRU sampling require $O(n^2)$ communication rounds, negating FALCON’s performance advantage over ML-DSA.
3. FALCON was not selected as NIST’s primary standard (ML-DSA was); it is a secondary standard under FIPS 206 (draft). Building threshold cryptography on a secondary standard adds adoption risk.

Quasar’s position. Neither threshold ML-DSA nor threshold FALCON are used. Ringtail fills the threshold PQ niche. Individual ML-DSA signatures provide non-threshold PQ identity binding, aggregated by Z-Chain into constant-size Groth16 proofs.

13 Conclusion

Quasar is the first production consensus protocol to require three independent cryptographic assumptions for finality. Safety holds if at least one layer remains secure. Liveness is guaranteed under partial synchrony with $f < n/3$ Byzantine validators. The protocol is implemented in Go,

deployed on Lux Network, and supported by machine-checked proofs in Lean 4 and symbolic verification in Tamarin.

Performance is practical: 69 ms full threshold rounds at $n = 100$ validators, 18.8 ns finality verification, and 20.26M messages per second on the ZAP transport layer. Epoch-based pruning bounds memory at 5.9 MB regardless of chain length. The test suite covers 2,160+ assertions across all consensus modes.

The quantum-layer design is deliberately conservative. We do not claim that BLS12-381 will resist quantum computers, nor that Ring-LWE or Module-LWE will survive the next two decades of cryptanalysis. We claim that requiring an adversary to break all three simultaneously—across three algebraically independent hardness assumptions—provides a margin of safety that no single-scheme approach can match.

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